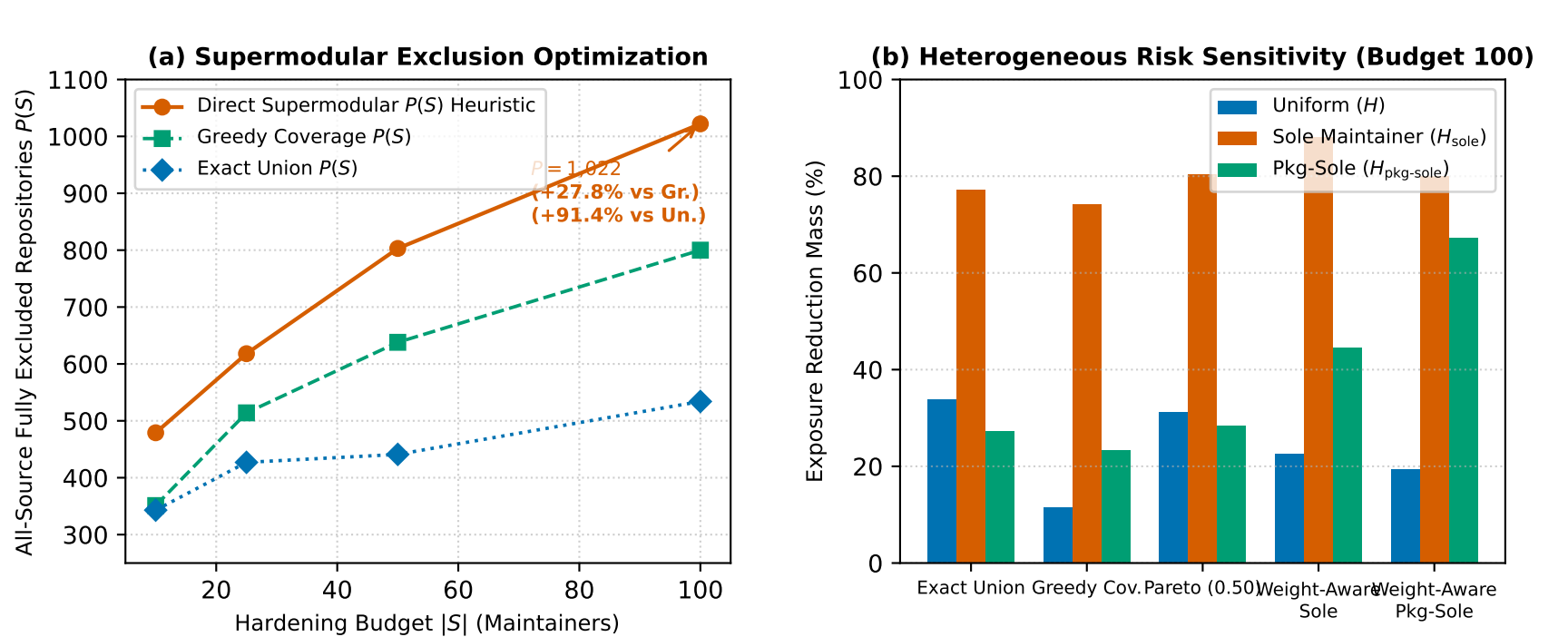
(a) Supermodular Exclusion Optimization
All-Source Fully Excluded Repositories $P(S)$
Hardening Budget $|S|$ (Maintainers)
Direct Supermodular $P(S)$ Heuristic
Greedy Coverage $P(S)$
Exact Union $P(S)$
$P = 1,022$
(+27.8% vs Gr.)
(+91.4% vs Un.)
1100
1000
900
800
700
600
500
400
300
20
40
60
80
100
(b) Heterogeneous Risk Sensitivity (Budget 100)
Exposure Reduction Mass (%)
Uniform (H)
Sole Maintainer (H_{sole})
Pkg-Sole ($H_{\text{pkg-sole}}$)
100
80
60
40
20
0
Exact Union
Greedy Cov.
Pareto (0.50)
Weight-Aware Sole
Weight-Aware Pkg-Sole